

Sapote–Mamey — Generic Deliverable Prompt Library (v1)

Reusable, parameterized prompts. Paste **GLOBAL GUARDS** once per session, then any G-block, filling {PARAMS}.

Readiness audit (against the v9.5.9 toolset)

prompt	driving tool	status	note
G1 BGC Atlas	tools/generate_bgc_report.py	READY	built this cycle (was missing); emits tier-coloured HTML
G2 Cross-Cohort Panel	build_figures.py --subset	GAP	--subset flag does not exist; use the filter→plot fallback below until added
G3 Strain Lead One-Pager	build_priority_leads.py → md/PDF	READY	tool yields the ranked leads; one-pager assembly is the narrow Sapote step
G4 Single-BGC Mode B	build_modeb_deepdiv.py --targets {S}:{B}	READY	deterministic; §1–§8
G5 Cross-Strain Table	add_xstrain_sheets.py	READY	confirm it covers {AXIS}; else per-strain tally fallback
G6 Priority-Leads Figure	build_priority_leads.py (+ figure)	PARTIAL	data ready; cohort-subset <i>figure</i> needs the same --subset as G2
G7 Layperson Guide	(Sapote LLM step)	READY	no tool; ranked + plain-language
G8 Manuscript Paragraph	(Sapote LLM step)	READY	every number sourced to a bank

prompt	driving tool	status	note
G9 Pangenome / Rarefaction	<code>build_pangenome.py</code> [<code>--replot</code>]	READY	families are anchor- based/approximate

Open gap (G2/G6 --subset). Until a `--subset {PRODUCT_TAG}/{COHORT_SUBSET}` flag is added to `build_figures.py` (or a `build_subset_panel.py` is created), produce these by: (1) deterministic filter+join of `bgc_data.json` to the tag/cohort into a locus CSV, (2) plot that CSV. The data step is reproducible; only the plotting wrapper is missing. (*Registered as DLV-008 on the HIVE board.*)

GLOBAL GUARDS (paste once per session)

- Build from **banked data only** (`merged_cohort/*.json`, `modeb_verdicts.csv`); never re-scan packages (skip `build_deep_data` / `build_bgc_markers`).
- **KCB = similarity anchor, not identity.** azoxy-crosslink, ~enediyne, ~halogenase are antiSMASH [**E-signals**], not structures.
- **Bioactivity = extract-level default** unless strain-specific assay data says otherwise.
- **AS verdicts are [EG]** (offline) until verified-literature-upgraded — tag them.
- **Corrected BGC count = Interior + $\frac{1}{2}$ · Edge + $\frac{1}{4}$ · Full-contig.**
- **Confidence tiers:** Confirmed · Predicted-functional · KCB-anchored · Candidate-novel (no anchor). Palette in `DELIVERABLE_INSTRUCTION_TEMPLATE.md`.
- **Any deliverable containing AS strains is PRIVATE** — no public release pending PI clearance. Don't surface AS strain IDs in public-facing text.
- Every locus shown carries: `Strain / Region · class · ~KCB anchor · edge-status · SARP y/n · ModeB verdict`.
- On finish: hand back the file + a 2–3 line note, and update the relevant HIVE Board row → DONE.

Parameter glossary: `{STRAIN}` · `{BGC}` region/NODE id · `{PRODUCT_TAG}` antiSMASH class tag · `{COHORT}` merged/SID/AS/habitat · `{STRAIN_SET}` comma list · `{AXIS}` feature axis · `{SECTION}` manuscript section.

G1 — BGC Atlas for {STRAIN}

Goal: browsable HTML atlas of `{STRAIN}`'s BGC inventory. **Type:** BGC Atlas (HTML). **Method:** `python tools/generate_bgc_atlas.py --strain {STRAIN} --banked-dir merged_cohort [--out {STRAIN}_BGC_Atlas[_PRIVATE].html]`.

Sapote step (narrow): bucket regions into bioactivity axes; unassignable → “Other / cryptic.” **Output:** header (strain/host/region+corrected count/strictness); one card per region (class · ~KCB anchor · kb · edge · tier · ModeB); legend; claim-safety footer. **Accept:** all regions · corrected count · tiers+legend · PRIVATE if AS.

G2 — Cross-Cohort Panel for {PRODUCT_TAG}

Goal: comparative figure of every {PRODUCT_TAG} locus in {COHORT}, grouped by host/habitat. **Method (until --subset lands):** filter `bgc_data.json` to {PRODUCT_TAG}, join `bgc_markers+tfbs_coupling+modeb_verdicts+strains.json` → locus CSV; plot lanes by habitat. **Do not infer a shared product** — KCB anchors may differ. Caption: the shared thread is the {PRODUCT_TAG} [E-signal]/machinery, not one compound. **Accept:** all tagged loci · habitat from `strains.json` · SARP+[EG] shown · PRIVATE if AS.

G3 — Strain Lead One-Pager for {STRAIN}

Goal: one-page ranked-lead + arsenal summary. **Method:** `build_priority_leads.py` rows for {STRAIN} → rank Class-A→C; Sapote writes 1-line rationale per lead; emit md/1-page PDF. **Accept:** ranked+tiered · rationale 1 line · claim-safety footer · PRIVATE if AS.

G4 — Single-BGC Mode B Deep Dive for {STRAIN}/{BGC}

Goal: §1–§8 dive on one cluster. **Method:** `python tools/build_modeb_deepdive.py --banked-dir merged_cohort --targets {STRAIN}:{BGC} --out {STRAIN}_{BGC}_ModeB.md`. **Accept:** all 8 sections · anchor flagged as similarity · verdict + next-step.

G5 — Cross-Strain Comparison Table for {STRAIN_SET} on {AXIS}

Goal: side-by-side {STRAIN_SET} × {AXIS} table. **Method:** `tools/add_xstrain_sheets.py` if it covers {AXIS}, else deterministic per-strain tallies; Sapote 2–3 sentence read-out. Flag **strictness confounds** if loose vs relaxed mixed. **Accept:** all strains · strictness flagged · 3-sentence note · PRIVATE if AS.

G6 — Priority-Leads Figure for {COHORT}

Goal: two-panel ranked-leads-over-field figure. **Method:** `build_priority_leads.py` for data; figure via `build_figures.py` (cohort-subset needs the G2 --subset fallback). Harmonize strictness before cross-cohort class-count claims. **Accept:** Class-A highlighted · counts match banks · PRIVATE if AS.

G7 — Layperson Ranked Guide for {STRAIN}

Goal: plain-language ranked cluster guide. **Method:** rank by novelty+tier+bioactivity; Sapote writes 2–4 jargon-free sentences/lead. **Accept:** ranked · jargon-free · confidence in plain terms · PRIVATE if AS.

G8 — Claim-Safe Manuscript Paragraph for {SECTION}

Goal: drop-in {SECTION} paragraph grounded in banked facts. **Method:** Sapote drafts 1 paragraph; every number tied to [bank: file/field]; flag any [EG]-dependent sentence. **Accept:** every number sourced · [EG] flagged · 1 paragraph.

G9 — Pangenome / Rarefaction for {COHORT}

Goal: pan-BGC-ome + rarefaction. **Method:** python tools/build_pangenome.py --banked-dir merged_cohort [--replot] --out fig_pangenome_{COHORT}.png. **Accept:** core/accessory/private counts · approximate-families caveat · saturation noted.

Adding a new generic prompt

Copy a block → G{n} - {Deliverable} for {PARAM}; list {PARAMS}; point Inputs at the banks; name the driving tool (or the Sapote judgment step); keep **Guards:** GLOBAL; register a HIVE board row and add it to the readiness audit.